

Moral Hazard I

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Readings: MSPC Chapter 3.1-3

“A moral hazard is where **one party is responsible for the interests of another, but has an incentive to put his or her own interests first**. The standard example is a worker with an incentive to shirk on the job. Financial examples include the following: I might sell you a financial product (e.g., a mortgage) knowing that it is not in your interests to buy it. I might pay myself excessive bonuses out of funds that I am managing on your behalf; or I might take risks that you then have to bear. Moral hazards such as these are a pervasive and inevitable feature of the financial system and of the economy more generally.”¹

¹Taken from Dowd, Kevin (2009): “Moral Hazard and the Financial Crisis,” Cato Journal 29(1), 141-166.

“... moral hazard occurs when insurance or policy inadvertently and self-defeatingly encourages people to behave in more risky ways.... Primary moral hazard is key to understanding risk during the COVID-19 pandemic, given that the costs associated with illness are not necessarily incurred by the person making risky choices.²

²<https://blog.petrieflom.law.harvard.edu/2020/09/03/moral-hazards-covid19-pandemic/>

Principal-agent relation with moral hazard

- One individual (agent A) acts on behalf of another individual (principal P) and is supposed to advance the principal's goals.
- A undertakes an action that affects P 's utility and this action is unobservable for P (**information asymmetry**).
 - Sometimes P may observe a “signal” (e.g. output) that is stochastically related to the agent's effort
- The agent and the principal have differing individual objectives (**conflict of interest**)
- The conflict of interest together with information asymmetry leads to problems of post-contractual opportunism

- The term “moral hazard” comes from the insurance industry, which differentiates between two types of risk.
- **“Objective Risk”**, is the unavoidable risk of a lightning strike, cancer or certain accidents, etc.
- **“Moral Hazard”**, is the part of a risk where the probability or the expected size of damage can be influenced by the insurance holder.
- Objective risk can be diversified through the financial market. risk attributable to moral hazard is harder.

- Fire insurance: The insurance holder may or may not spend time and money for protective measures that reduce the probability of damage.
- Automotive insurance: The insurance holder may or may not drive carefully.
- Health insurance: The insurance holder may or may not adopt a healthy lifestyle or attend medical checks on a regular basis.

Overview

Questions:

- Consequences of asymmetric information on efficiency of allocation?
- Design of **incentive schemes** that induce the agent to take a certain action. What does the optimal incentive scheme look like?

Overview:

- Formal principal-agent model:
- First best contract (effort observable)
- Second best contract (effort unobservable)

- We return to the same framework (with minor changes: e.g. fewer dimensions in action space and fewer states of nature) as we covered in our first week (benchmark model)
- A classic problem of moral hazard is the conflict between an employer and an employee or maybe an owner and the manager of a company.
- **Asymmetric Information.** The owner cannot directly observe or evaluate the manager's actions.
 - In an alternative setting she may observe the manager but she may not know on what information the manager bases his decisions or what is best-effort choice (moral hazard with hidden information, more later).
- **Conflict of Interests.** The owner wants the manager to increase the value of the company. On the contrary, the manager wishes to maximize her utility, i.e., work less hard, enjoy perks (representative office, corporate jet, pointless firm mergers).

- We build a simple model to analyze the owner-manager conflict.
- Our goal is to find an incentive contract that maximizes the owner's utility under the constraint that the manager exerts a certain level of effort.

- We call the owner of the company the “principal” P , and the manager the “agent” A .
- The possible profit levels (or “states of the world”) are $x \in \{x_s, x_f\}$, where $x_f < x_s$. (Success, Failure)
- A chooses between actions, $e^l = 0$ and $e^h = 1$, where e^l represents low effort and e^h high effort.
- The costs of action for A are given by $C(e^h) = c > C(e^l) = 0$.
- If A does not work for P , he gets his reservation utility $\bar{U}$.

Utility Agent:

- A is risk-averse with utility function $U(w, e) = v(w) - C(e)$, where $v' > 0$ and $v'' < 0$.
- $v(\cdot)$ is the agent's Bernoulli utility function
- inverse utility function $h = v^{-1}$ strictly increasing and convex
- reservation utility of agent: $\bar{U}$

Utility Principal:

- P is risk-neutral and has the utility function $\pi = x - w$, where w is the wage P pays to A .
- why is it plausible that the principal is risk neutral?

- A's action determines the probability distribution over the company's profit x . In particular denote by $p_s(e)$ the probability of a successful outcome x_s :

$$1 > p_s(e = 1) > p_s(e = 0) > 0.$$

Since $p_s(e) + p_f(e) = 1$, we also have

$$0 < p_f(e = 1) < p_f(e = 0) < 1$$

- every state occurs with positive probability under each effort. Why is this important?

The sequence of events is as follows:

- ① P designs and offers an incentive contract $w(x)$ or $w(e, x)$.
- ② A decides whether to accept or reject P 's contract. If A rejects the contract, P gets 0 and A gets $\bar{U}$. Otherwise, the game continues.
- ③ A chooses his effort e^i (non-verifiable in the second best).
- ④ The profit x is realized, P gets $x - w(x)$ and A gets $w(x) - C(e^i)$.

- Suppose that there is no asymmetric information. Then P can implicitly determine A 's action in the contract (for example, by imposing a huge fine on A if he deviates to another action).
- Which incentive contract $w(e, x)$ then maximizes P 's expected profits? We answer this question by going through two steps.
- First, we determine for each possible action the **optimal incentive contract** that implements the action at lowest costs.
- Second, we choose **the action that P would like to implement** (given the minimal costs from the first step).
 - why two steps?

Implementing e^h

- We start with e^h .
- The optimal contract solves the problem

$$\max_{w_s, w_f} p_s(1)(x_s - w_s) + [1 - p_s(1)](x_f - w_f)$$

subject to

$$p_s(1)v(w_s) + (1 - p_s(1))v(w_f) - c \geq \bar{U}.$$

- In the optimum, PC must be binding (why?).
- Note that $x_i p_i^h$ is a constant in the maximization problem over w_i and hence can be neglected. Furthermore, note that $\max_w \sum_i -w_i p_i^h$ is equivalent to $\min_w \sum_i w_i p_i^h$.
- The constraint is called the **participation constraint** (or individual rationality constraint)

- The Lagrangian function is

$$\begin{aligned}\mathcal{L}(w_s, w_f, \lambda) &= p_s(1)(x_s - w_s) + [1 - p_s(1)](x_f - w_f) \\ &+ \lambda[p_s(1)v(w_s) + (1 - p_s(1))v(w_f) - c - \bar{U}]\end{aligned}$$

- We treat the IR constraint as binding for now, why?
- The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial w_i} = -p_i(1) + \lambda p_i(1)v'(w_i^*) = 0 \text{ for } i = s, f$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = p_s(1)v(w_s) + (1 - p_s(1))v(w_f) - c - \bar{U} = 0.$$

- The first-order conditions characterise the optimal contract. From the first equality we obtain

$$\frac{1}{v'(w_i)} = \lambda \text{ for all } i \in \{f, s\}.$$

- This means $w_s = w_f = \bar{w}(e^h)$, where $\bar{w}(e^h)$ is A 's wage if he chooses e^h .
 - i.e. in the first best contract, P is not using x_i in determining pay, he instead directly uses and contracts upon e_i
- A constant wage implies that A is fully insured against the risk inherent in the bilateral relationship.

- From the IR constraint we obtain

$$v(\bar{w}(e^h)) - c = \bar{U}.$$

- We can rewrite this as

$$\bar{w}(e^h) = v^{-1}(\bar{U} + c) = h(\bar{U} + c)$$

where $h = v^{-1}$.

- Hence, A is compensated for his outside option and gets his reservation value.

Thus, we obtain the implementation costs/wage for effort $\hat{e} = e^h$:

$$w(\hat{e}) = \begin{cases} \bar{w} = h(\bar{U} + c(\hat{e})) & \text{if } e = \hat{e} \\ \bar{w} - f & \text{if } e \neq \hat{e} \end{cases}$$

The fine f ensures that agent indeed takes the desired effort. That is, it is chosen such that a deviation from $\hat{e}$ does not pay off:

$$v(\bar{w} - f) - c(e) < v(\bar{w}) - c(\hat{e})$$

or in our case

$$v(\bar{w} - f) < v(\bar{w}) - c(e^h)$$

This fine can be given since effort is observable and verifiable.

Implementing e^l and comparing

- As with high effort, we can derive the optimal incentive contract that implements low effort e^l . Again, we obtain a fixed wage $w(e_l) = h(\bar{U})$. (why?).
- Lastly, P chooses the contract that offers her the highest expected profit:

$$\begin{aligned}
 & p_s(1)x_s + (1 - p_s(1))x_f - h(\bar{U} + c(\hat{e})) \\
 \geq & p_s(0)x_s + (1 - p_s(0))x_f - h(\bar{U})
 \end{aligned}$$

Efficiency

- Note that under the optimal contract, both the allocation of risk (A has full insurance while P absorbs all the risk) and effort incentives (A chooses the optimal action) are efficient.
 - A does not like risk hence would be willing to accept lower wage for less risk.
 - gains from trade are exhausted. P does not want A to work more.
 - “one tool (wage) to solve one target (agent participation)” ... later wage also has to induce particular effort...

Contractual assumptions

- Effort is unobservable to the principal
 - wage scheme cannot condition on effort
 - first best contract potentially not incentive compatible (next slide)
- Output is observable and verifiable
 - wage scheme can condition on output
- $\Rightarrow \mathbf{w} = (w(x_s), w(x_f))$.
Let $w(x_i) = w_i$.

- Under a fixed wage, A would always choose e^l .
 - First best solution with high effort is not implementable (nor incentive compatible). Why?
 - What does the optimal incentive contract look like under asymmetric information?
- Intuitively a contract that implements e^h has to pay more to A in a state of the world that is more likely after high effort e^h than after low effort e^l .
- expect that:
 - 1 wage varies
 - 2 contract is more costly for P

- Again, we proceed in two steps.
- First, we determine for each possible action the optimal incentive contract that implements the action at lowest costs.
- Second, we choose the action that maximizes P 's profit.

- We start with e^h .
- P's maximization problem is

$$\max_{w_s, w_f} p_s(1)(x_s - w_s) + [1 - p_s(1)](x_f - w_f)$$

subject to the **participation constraint (PC)**

$$p_s(1)v(w_s) + (1 - p_s(1))v(w_f) - c \geq \bar{U}.$$

AND the **incentive compatibility constraint (IC)**

$$p_s(1)v(w_s) + (1 - p_s(1))v(w_f) - c \geq p_s(0)v(w_s) + (1 - p_s(0))v(w_f).$$

- What does the IC constraint secure?

- Intuitively the IC constraint ensures that the agent does as we “expect” him to do.
 - IR: A participates
 - IC: A performs the right action
- With more effort levels more IC constraints.
 - hence we may also expect there to be situations where some of the constraints become redundant.
 - it is a major simplification of these problems if this happens and we know/realize it :-)

- ① From the IC constraint we immediately also see that a fixed wage scheme (and $c > 0$) is not incentive compatible when P wants to implement e^h .

$$\begin{aligned} & [p_s(1) - p_s(0)]v(w_s) + [p_s(0) - p_s(1)]v(w_f) \geq c \\ \Leftrightarrow & [p_s(1) - p_s(0)][v(w_s) - v(w_f)] \geq c \end{aligned}$$

Under a fixed wage scheme $[v(w_s) - v(w_f)] = 0$ and hence the inequality is not satisfied.

- ② It may also make sense to expect the IC to be binding, as else P could offer lower wages (w_s) which would still make A perform e_h . This of course assumes that the IR constraint is still fulfilled (binding or non-binding).

- The Lagrangian function becomes

$$\begin{aligned}\mathcal{L}(w_s, w_f, \lambda, \mu) &= p_s(1)(x_s - w_s) + [1 - p_s(1)](x_f - w_f) \\ &+ \lambda[[p_s(1) - p_s(0)][v(w_s) - v(w_f)] - c] \\ &+ \mu[p_s(1)v(w_s) + (1 - p_s(1))v(w_f) - c - \bar{U}].\end{aligned}$$

- One can show that the solution to the first order conditions is indeed a maximum of the Lagrangian (see MSPC).

The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial w_s} = -p_s(1) + \lambda(p_s(1) - p_s(0))v'(w_s^*) + \mu p_s(1)v'(w_s^*) = 0 \quad (1)$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial w_f} &= -(1 - p_s(1)) \\ &\quad - \lambda(p_s(1) - p_s(0))v'(w_f^*) + \mu(1 - p_s(1))v'(w_f^*) = 0 \end{aligned} \quad (2)$$

Complementary slackness conditions:

$$\lambda((p_s(1) - p_s(0))[v(w_s^*) - v(w_f^*)] - c) = 0$$

$$\mu[(p_s(1)v(w_s^*) + (1 - p_s(1))v(w_f^*) - c - \bar{U}] = 0$$

See appendix of MSPC for Kuhn-Tucker theory.

- Under the optimal contract, the PC must be binding (why?):
- If not, a participation tax could be charged in the market. I.e. we could change w_f and w_s such that IC is still binding but IR would now also bind.

$$(p_s(1) - p_s(0))[v(w_s) - v(w_f)] \geq c$$

- IC mainly says something about the relative awards.

See that IR is binding

Setting $\mu = 0$ in (2) leads to a contradiction.

IR binding

Hence

$$p_s(1)v(w_s^*) + (1 - p_s(1))v(w_f^*) - c = \bar{U}$$

- A is indifferent between accepting or rejecting the contract. He is compensated for his outside option and effort costs. The expected wage payment to A is larger under asymmetric than under symmetric information (why?).

IC constraint

Under the optimal contract, the IC also binds: (why?)

$$[p_s(1) - p_s(0)][v(w_s^*) - v(w_f^*)] = c$$

- Hence, at the optimum A is indifferent between high and low effort. In equilibrium (if P decides to implement the high effort contract), he will choose high effort.

- The first-order conditions characterize the optimal contract.

We must have

$$\frac{1}{v'(w_s^*)} = \mu + \lambda \left[1 - \frac{p_s(0)}{p_s(1)} \right] \quad (3)$$

$$\frac{1}{v'(w_f^*)} = \mu + \lambda \left[1 - \frac{1 - p_s(0)}{1 - p_s(1)} \right] \quad (4)$$

- Since the Lagrange multipliers are positive constants, this implies that the wage is not constant, but moves with the likelihood-ratio $p_s(0)/p_s(1)$.
 - as $p_s(1)$ increases the RHS in (3) becomes larger (smaller in (4)). This implies that w_s^* (w_f^*) becomes larger (smaller).

Likelihood ratios

The wage in a state of the world increases if and only if $\frac{p_s(0)}{p_s(1)}$ decreases.

Why?

Interpretation in principal agent context:

- $\frac{p_s(0)}{p_s(1)}$ small: very likely that agent worked hard compared to shirked when we observe state i .
- $\frac{p_s(0)}{p_s(1)}$ large: very unlikely that agent worked hard compared to shirked when we observe state i .
- Principal wants to put rewards (high wages) in situations where it is relatively more likely that high effort was exerted and punish in situations where it is relatively less likely.

Low effort

- How does the optimal contract look if P wants to implement e^l ?
- What are the implementation costs?
- To save on implementation costs, the principal will sometimes implement low effort in the second best, while implementing high effort is chosen in the first best (and efficient)

- When e_h is chosen under the first best the second best contract leads to an inefficient allocation compared to the first best because of either:
 - 1 risk sharing: although A is risk-averse and P risk-neutral, A has to bear some risk.
 - Contracts with a fixed wage does not work: P needs to rely on realized output as “indicators” for whether A actually exerted high effort.
 - P tries to motivate A to high effort through a share in the risky profits.
 - But as A dislike risk a risk premium needs to be paid (costly). Therefore P also tries to reduce the risk premium she has to pay A .
 - 2 inefficient effort: when e_h is chosen under the first best but not under the second best there is an inefficient level of effort in SB. A would not be worse off by the first best contract but he cannot credibly commit to deliver e_h .

Implications - Efficiency

The contracting outcome under asymmetric information is called second-best solution because it needs to satisfy the additional incentive constraint and therefore does not always correspond to the first-best solution.

As a consequence of this additional constraint, the outcome under asymmetric information may fail to be Pareto optimal – leading to either inefficient risk sharing or to an inefficiently low effort level.

- If high effort is implemented risk-sharing is inefficient since the risk-averse A is not fully insured by the risk-neutral P ;
 - P prefers first best, A is indifferent (gets $\bar{U}$). Hence they would be willing to trade (can make one better without “hurting” the other) BUT due to post contractual opportunism the trade is not feasible.
- Low effort might be implemented even though high effort is socially optimal.

Additional Information

Suppose that P can observe two signals, x and y . For example, x may be the companies profit and y the profit of P 's rivals.

Should the optimal contract condition on both signals?

The answer is yes if and only if the second signal contains additional information about A 's action. The more information P has about A 's action, the smaller is the risk-premium she has to pay to A .

Risk-neutral Agent

What is the optimal contract if P cannot observe action, but both P and A are risk-neutral?

Now it does not matter for efficiency who bears the risk.

The following incentive contract implements the first-best allocation:

$$w_i = x_i - K$$

where K is chosen so that A gets his reservation utility.

- Interpretation: P sells the company to A at price K .
- This solution of course requires that A is not wealth constrained.